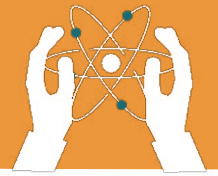


UNIT 8

SOUND PROPERTIES AND NOISE

Learning objectives:

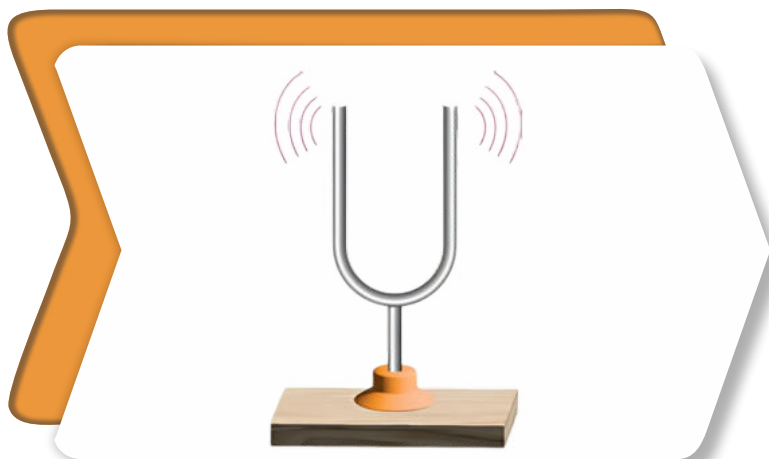
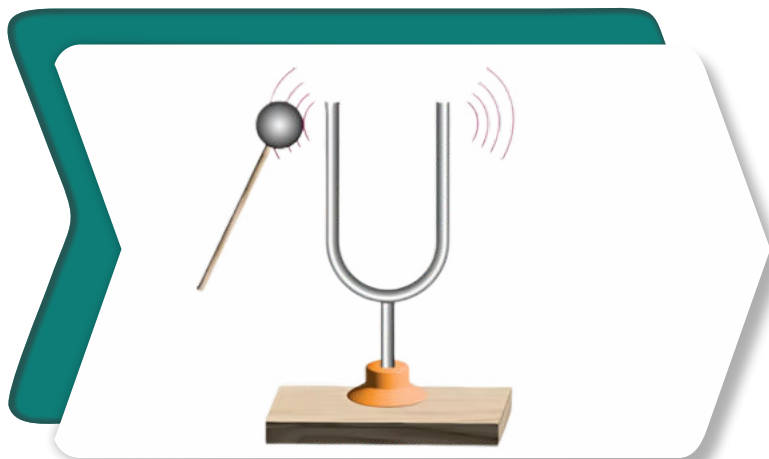
- Understand that sound is created by vibrations and travels through different mediums.
- Learn about the characteristics of sound: Pitch, Volume, and Frequency.
- Recognize the effects of noise pollution and ways to reduce it.

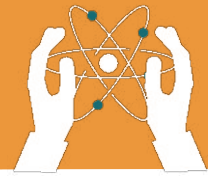


Sound is a form of energy. It is important part of our daily life. We hear sounds everywhere birds chirping, people talking, and even the wind blowing. In this chapter, we will learn about the properties of sound, how it travels, and the effects of noise pollution on our health.

What happens when you hit the drum?

When we hit an object with another object, it produces vibrations, which produce sound or sound waves that travel through the air or other materials. These waves reach our ears, and we hear the sound.





SOUND PROPERTIES AND NOISE

How Does Sound Travel?

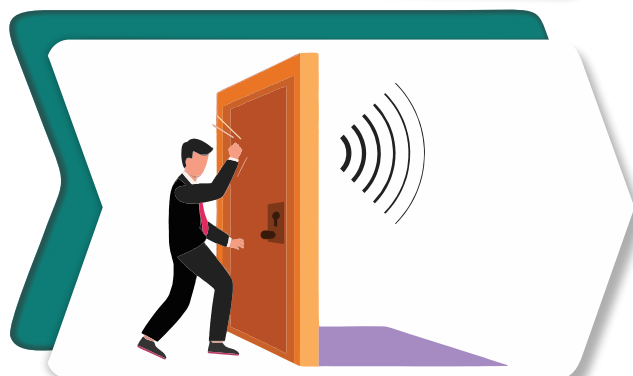
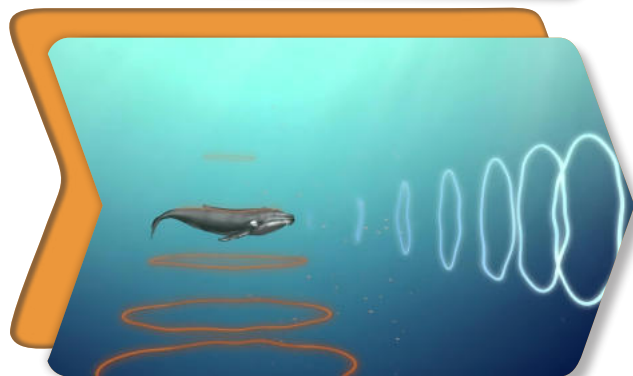
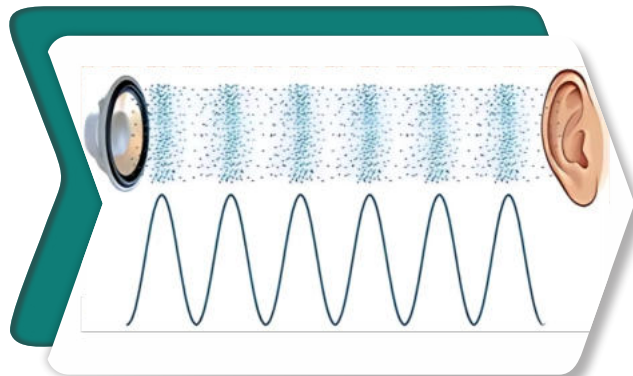
Sound travels in **waves** and needs a medium to pass through. This can be air, water, or even solids.

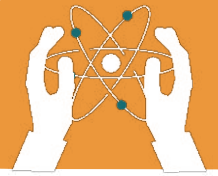
1. Through Air: Sound travels to our ears through the air.

2. Through Water: Sound moves faster in water than in air.

3. Through Solids: Sound moves fastest through solids, like metal or wood, because their particles are tightly packed.

Example: You can hear a knock on a wooden table because the sound travels through the solid wood.





Characteristics of Sound

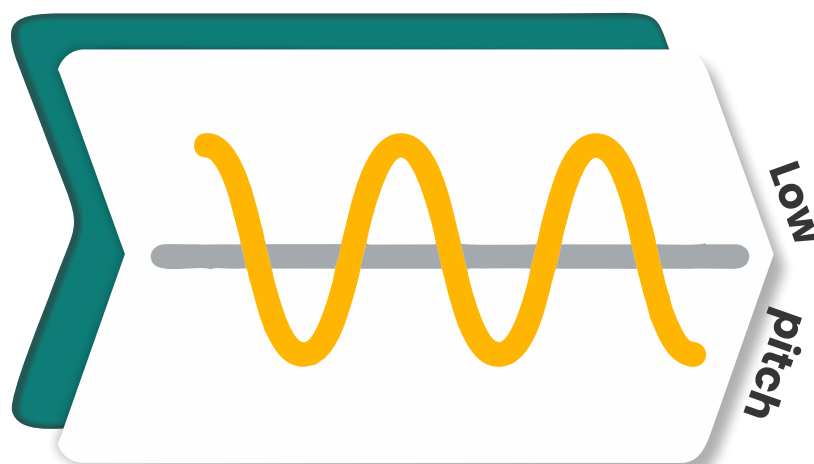
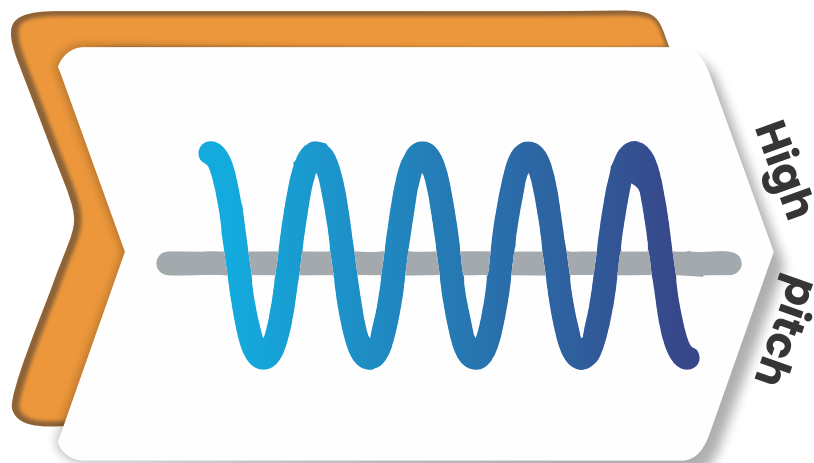
1. Pitch

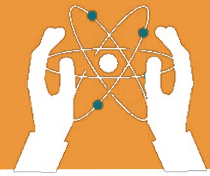
Pitch tells us how high or low a sound is.

High Pitch: Sounds like a whistle or bird chirping (high-frequency waves).

Low Pitch: Sounds like a drumbeat or a lion's roar (low-frequency waves).

Example: A flute produces high-pitched sounds, while a tuba produces low-pitched sounds.





SOUND PROPERTIES AND NOISE

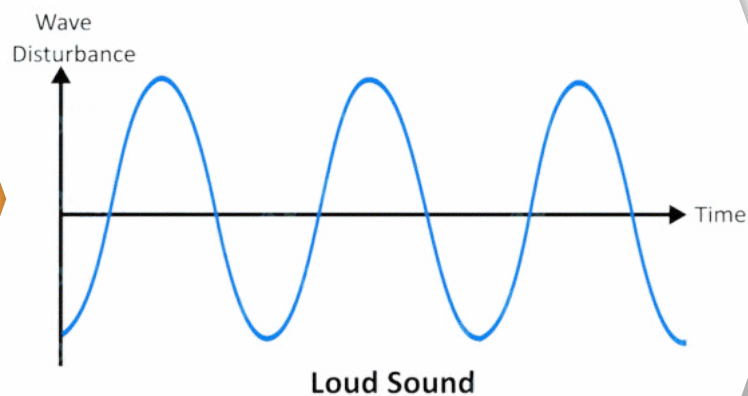
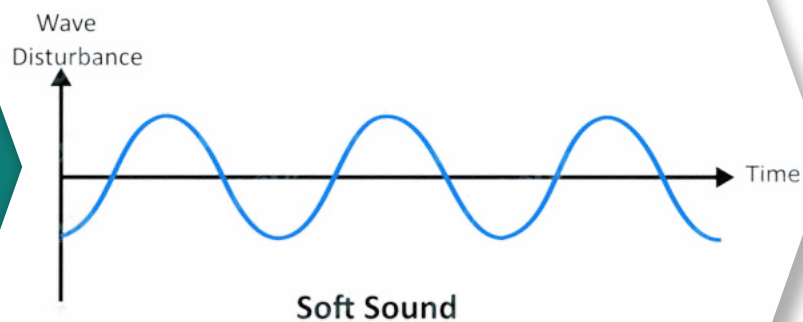
2. Volume

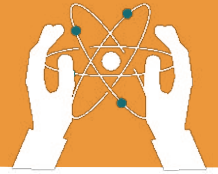
Volume tells us how loud or soft a sound is.

Loud Sound: High amplitude (e.g., a firecracker).

Soft Sound: Low amplitude (e.g., a whisper).

Example: Turning the volume up on your phone makes the sound louder.





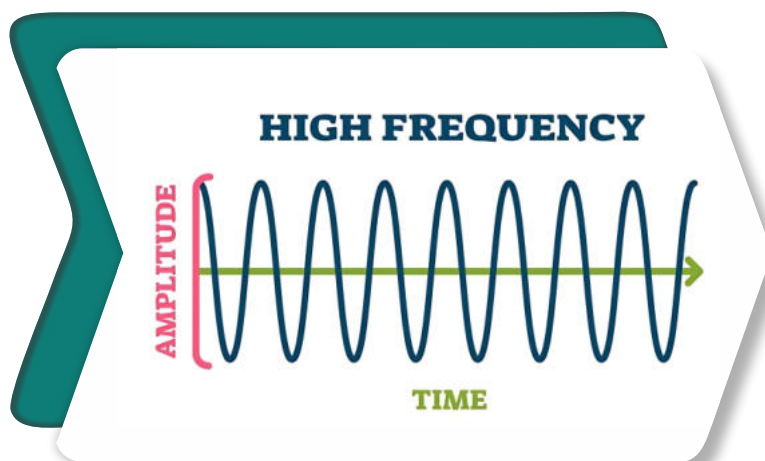
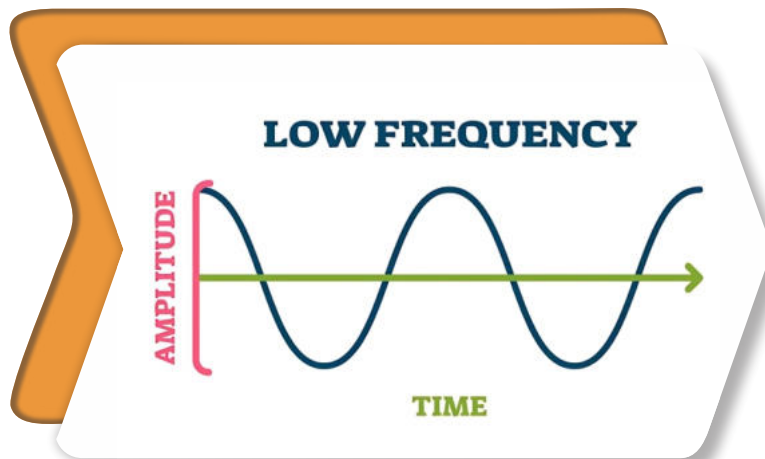
3. Frequency

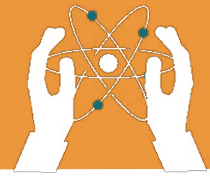
Frequency tells us how fast the sound waves vibrate.
Measured in Hertz (Hz).

High Frequency: Many sound waves per second, creating a high pitch.

Low Frequency: Fewer sound waves per second, creating a low pitch.

Example: A mosquito buzz has high frequency, while a lion's roar has low frequency.



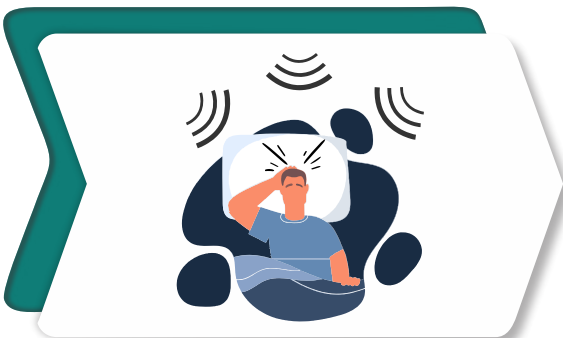


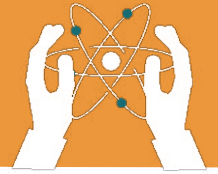
Noise Pollution

Not all sounds are pleasant. Loud, unwanted sounds are called **noise**. Examples include honking cars, loudspeakers, and factory machines. Noise pollution can disturb us and cause harm to our health.

Effects of Noise Pollution

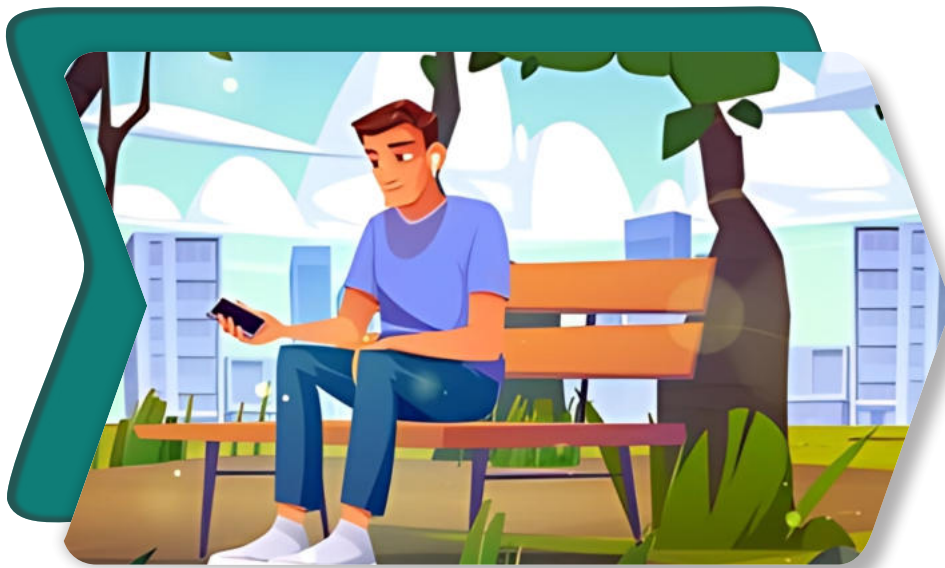
1. Causes stress and irritation.
2. Causes global warming.
3. Makes it hard to focus on work or studies.
4. Can harm hearing if the noise is too loud.
5. Affects sleep, making us tired and cranky.

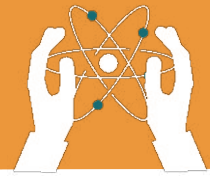




How to Reduce Noise Pollution

1. Avoid playing loud music.
2. Use earplugs in noisy places.
3. Plant trees, as they absorb noise.
4. Follow "no honking" signs in traffic.



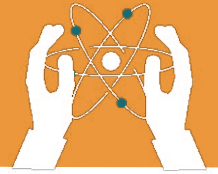


SOUND PROPERTIES AND NOISE

Fill in the blanks.

1. Sound is created by _____.
2. Sound travels in _____.
3. The _____ of a sound tells us how loud it is.
4. _____ is a type of unwanted sound.
5. High-pitched sounds have _____ frequency waves.
6. Loud sounds have _____ amplitude.
7. Noise pollution can cause _____ and hearing damage.
8. Planting _____ can help reduce noise pollution.

SOUND PROPERTIES AND NOISE



Write “T” for True or “F” for False.

1. Sound travels faster in water than in air.

2. A drum produces a high-pitched sound.

3. Noise pollution can make it difficult to sleep.

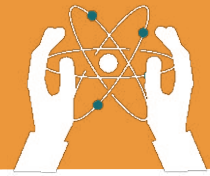
4. High-frequency sound waves create low-pitched sounds.

5. Sound travels in waves.

6. Soft sounds have high amplitude.

7. Loud music reduces noise pollution.

8. Vibrations create sound.



Choose the correct answer.

1. What creates sound?

- Light waves
 - Vibrations
 - Heat waves
 - Trees
-

2. In which medium does sound travel the fastest?

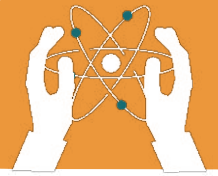
- Air
 - Water
 - Solids
 - Space
-

3. Volume tells us:

- How loud or soft a sound is
 - How fast sound waves vibrate
 - How sound travels
 - Makes trees grow faster
-

4. What happens if noise pollution is too loud?

- It damages hearing
 - It makes trees grow faster
 - It creates vibrations
 - It helps people sleep better
-



5. What is a good way to reduce noise pollution?

- Playing loud music
 - Honking in traffic
 - Planting trees
 - Increasing factory noise
-

6. Which of the following is a property of sound?

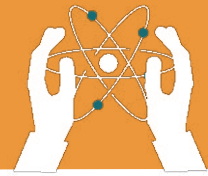
- Sound is visible
 - Sound travels through space
 - Sound is created by vibrations
 - Sound travels faster in air than in solids
-

7. What type of waves does sound travel in?

- Heat waves
 - Light waves
 - Sound waves
 - Water waves
-

8. Frequency is measured in:

- Kilograms
 - Hertz
 - Metres
 - Seconds
-



SOUND PROPERTIES AND NOISE

Answer the following questions.

Q1. How sound can be produced?

Q2. How does sound travel through solids, liquids, and gases?

Q3. Define pitch.

Q4. What is the difference between loud and soft sounds?

Q5. What is noise pollution?

Q6. How can we reduce noise pollution?

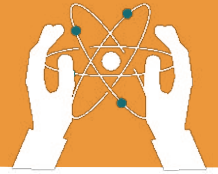
Q7. Name two effects of noise pollution on humans.

Q8. Explain the characteristics of sound: Pitch, Volume, and Frequency.



Critical Thinking

How can you protect your ear in an area where construction is going on?

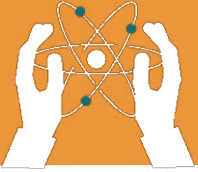


Create Vibrations to Make Sound



ACTIVITY

- **Materials Needed:** A rubber band and an empty tissue box or a small cardboard box.
- **Steps:**
 1. Stretch the rubber band around the tissue box.
 2. Pluck the rubber band with your fingers and listen to the sound.
 3. Notice how the rubber band vibrates to create sound waves.
 4. Try plucking the rubber band harder or softer to observe how the sound changes.



SOUND PROPERTIES AND NOISE

Are these sounds high or low? Circle the correct option.



High Both Low



High Both Low



High Both Low